Solving Ax = b: Row Reduced Form R

Now the client wants a specific target. Four short steps on one simplified matrix $[R \mid d]$ tell you whether it can be hit, give one portfolio that hits it, and then every portfolio that does.

PART 1 Index

Note 2 of the pair: $Ax = b$, solving for a specific target. (Note 1 is Lesson 7, $Ax = 0$.)

1. **Conditions and important concepts**
   12 short definitions: [A | b], [R | d], zero rows, contradiction, free variables, the three possibilities, particular and complete solutions, rank conditions.
2. **Procedure for Ax = b**
   The 7 steps, in order.
3. **2 × 2 examples**
   Exactly one, infinitely many, none.
4. **3 × 3 examples**
   Exactly one, infinitely many, none.
5. **Final decision tree**
   Contradiction? Free variables? The answer.
6. **Connection between Ax = 0 and Ax = b**
   Every solution = one solution + a null space solution. With a target tester.
7. **Final memory table**
   Ax = 0 and Ax = b side by side.

METHODS IN THIS LESSON: Augmented matrix [A | b] · [A | b] → [R | d] · Contradiction test · Particular solution · Complete solution $x_p + x_n$ · Rank conditions

PART 2 Prerequisites

Click any link to review it, then come back.

- Note 1: Ax = 0 (Lesson 7): R, pivots, free variables, special solutions, N.
- Ax = b is solvable exactly when b is in C(A) (Lesson 6).

PART 3

Explanation

THE FUNDS USED IN THE EXAMPLES

Nifty $(1, 12, 1)$, Nifty ETF $(1, 12, 1)$, bonds $(1, 7, 0)$, balanced $(1, 9.5, 0.5)$, midcap $(1, 15, 1.2)$: what ₹1 lakh of each does to (budget, return, exposure). A target asks for a budget (₹ lakh), a return (₹ thousand) and an exposure.

1. Conditions and important concepts

1. What is Ax = b?

$Ax = b$ means:

$$\boxed{\text{Find the portfolio } x \text{ that produces the specific target } b.}$$

2. What is b?

b is the **target**. For example:

$$b = \begin{bmatrix} B \\ R \\ E \end{bmatrix} = \begin{bmatrix} 10 \\ 105 \\ 7 \end{bmatrix} \qquad \begin{aligned} B &= \text{budget ₹10 lakh} \\ R &= \text{return ₹105 thousand} \\ E &= \text{exposure } 7 \end{aligned}$$

3. Augmented matrix [A | b]

METHOD · AUGMENTED MATRIX [A | B]

Attach the target b to A as an extra column. The vertical line separates the matrix (left) from the target (right).

For Nifty, bonds, balanced and the target above:

$$[\,A \mid b\,] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 7 & 9.5 & 105 \\ 1 & 0 & 0.5 & 7 \end{array}\right]$$

4. What happens to [A | b]?

> **METHOD · ROW REDUCE [A | B] → [R | D]**
>
> Row reduce the **entire** augmented matrix. The same row operations are applied to A **and** to b. The left part becomes R, exactly as in Note 1.

The row operations are Note 1's (3 × 3 example). Here is what each one does to the target column:

Row operation	Target column
Row 2 $-$ 12$\times$ row 1	$105 - 12(10) = -15$
Row 3 $-$ 1$\times$ row 1	$7 - 10 = -3$
Row 3 $-$ 0.2$\times$ row 2	$-3 - 0.2(-15) = -3 + 3 = 0$
Row 2 $\div(-5)$	$-15/(-5) = 3$
Row 1 $-$ 1$\times$ row 2	$10 - 3 = 7$

5. What is d?

After row reduction, the original target b becomes a simplified target column called d:

$$[R \mid d] = \left[\begin{array}{ccc|c} 1 & 0 & 0.5 & 7 \\ 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right] \qquad d = \begin{bmatrix} 7 \\ 3 \\ 0 \end{bmatrix}$$

6. Zero row

A zero row looks like $[\,0 \;\; 0 \;\; 0 \mid \text{something}\,]$. There are **two** possibilities.

Condition 1: $[\,0 \;\; 0 \;\; 0 \mid 0\,]$. This means

$$\boxed{0 = 0}$$

This is fine. There is **no contradiction**.

Condition 2: $[\,0 \;\; 0 \;\; 0 \mid \text{non-zero}\,]$. For example (target $(10, 105, 5)$, see the 3 × 3 examples):

$$\boxed{0 = -2}$$

This is impossible: whatever x_1, x_2, x_3 are, the left side is always zero. Therefore **no solution**.

7. Contradiction condition

> **RULE · CONTRADICTION TEST**
>
> If after row reduction we get a row saying $0 = \text{non-zero}$, then there is **no solution**.

8. Free variable condition

If there are **no contradictions**:

No free variables $\rightarrow$ exactly ONE solution

At least one free variable $\rightarrow$ INFIN

9. The three possibilities

$$\boxed{\begin{array}{c}\text{Contradiction} \rightarrow \text{NO SOLUTION} \\ \text{No contradiction} + \text{no free variables} \rightarrow \text{EXACTLY ONE SOLUTION} \\ \text{No contradiction} + \text{free variables} \rightarrow \text{INFINITELY MANY SOLUTIONS}\end{array}}$$

10. Particular solution

> **METHOD · PARTICULAR SOLUTION X_P**
>
> A particular solution is simply **one** solution that satisfies $Ax = b$. To find an easy one: set all free variables $= 0$, then solve for the pivot variables (each pivot variable $=$ its row's d).

For the $[R \mid d]$ above: free $x_3 = 0$, so $x_1 = 7$, $x_2 = 3$: $x_p = (7, 3, 0)$, ₹7 lakh Nifty and ₹3 lakh bonds.

11. Complete solution

Once we have one particular solution x_p, we can add any null space solution x_n:

$$\boxed{x = x_p + x_n}$$

Why? Because $Ax_p = b$ and $Ax_n = 0$. Therefore

$$A(x_p + x_n) = Ax_p + Ax_n = b + 0 = b.$$

> **RULE · COMPLETE SOLUTION**
>
> Every solution $=$ one solution $+$ a null space solution. (And nothing else: if x is any solution, $x - x_p$ gives $b - b = 0$, so it is a null space solution.)

For the example: $x_n = t\,(-0.5, -0.5, 1)$ from Note 1, so $x = (7, 3, 0) + t\,(-0.5, -0.5, 1)$.

12. Rank conditions

Suppose A has m rows, n columns and rank r. Then:

$$\text{number of zero rows} = m - r \qquad \text{number of free variables} = n - r$$

RULE · RANK CONDITIONS

If $r = m$: no zero rows, so never a contradiction: every target is reachable.
If $r = n$: no free variables: at most one solution.
If $r = m = n$: no zero rows **and** no free variables: exactly one solution for every target.

Funds (3 rules)	m	n	r	What it means
Nifty, bonds, midcap	3	3	3	$r = m = n$: exactly one solution for every target
Nifty, bonds	3	2	2	$r = n$: at most one solution
Nifty, bonds, midcap, balanced	3	4	3	$r = m$: every target reachable (infinitely many ways)
Nifty, bonds, balanced	3	3	2	neither: none or infinitely many

2. Procedure for Ax = b

Step	What to do
1	Build $[A \mid b]$.
2	Row reduce: $[A \mid b] \to [R \mid d]$.
3	Check the zero rows. $0 = \text{non-zero} \to$ **no solution** (stop). $0 = 0 \to$ continue.
4	Find the pivot and free variables.
5	No free variables → **exactly one** solution. Free variables → **infinitely many**.
6	To find **one** solution: set all free variables $= 0$ and solve for the pivot variables.
7	To find **every** solution: keep the free variables arbitrary (equivalently, add the null space solutions).

3. 2 × 2 examples

Budget and return only.

Example A: exactly one solution

Nifty and bonds, target $(10, 105)$:

$$\left[\begin{array}{cc|c} 1 & 1 & 10 \\ 12 & 7 & 105 \end{array}\right]$$

- Row 2 $-$ 12$\times$ row 1: $(0,\ 7 - 12 = -5 \mid 105 - 120 = -15)$.
- Row 2 $\div(-5)$: $(0, 1 \mid 3)$.
- Row 1 $-$ row 2: $(1, 0 \mid 10 - 3 = 7)$.

$$[R \mid d] = \left[\begin{array}{cc|c} 1 & 0 & 7 \\ 0 & 1 & 3 \end{array}\right]$$

No contradictions. No free variables. Therefore **exactly one solution**:

$$\boxed{x_1 = 7,\ \ x_2 = 3} \qquad x = \begin{bmatrix} 7 \\ 3 \end{bmatrix}$$

Check: $7 + 3 = 10$ ✓, $84 + 21 = 105$ ✓.

Example B: infinitely many solutions

Nifty and its twin ETF, target $(10, 120)$:

$$\left[\begin{array}{cc|c} 1 & 1 & 10 \\ 12 & 12 & 120 \end{array}\right] \xrightarrow{\text{row } 2 - 12 \times \text{row } 1} \left[\begin{array}{cc|c} 1 & 1 & 10 \\ 0 & 0 & 0 \end{array}\right]$$

(Row 2: $12 - 12 = 0, 12 - 12 = 0, 120 - 120 = 0$.) There is no contradiction ($0 = 0$). But x_2 is free. Therefore **infinitely many solutions**.

Let $x_2 = c$. Then $x_1 + c = 10$, so $x_1 = 10 - c$:

$$\boxed{x = \begin{bmatrix} 10 - c \\ c \end{bmatrix}} \qquad \text{any split of ₹10 lakh between the twins}$$

Example C: no solution

The twins again, but target $(10, 105)$:

$$\left[\begin{array}{cc|c} 1 & 1 & 10 \\ 12 & 12 & 105 \end{array}\right] \xrightarrow{\text{row } 2 - 12 \times \text{row } 1} \left[\begin{array}{cc|c} 1 & 1 & 10 \\ 0 & 0 & -15 \end{array}\right]$$

The second row says

$$\boxed{0 = -15}$$

Impossible. Therefore **no solution**. (Two copies of a 12% fund can't earn 10.5%.)

4. 3 × 3 examples

Example A: exactly one solution

Nifty, bonds, midcap, target $(10, 111, 7.4)$:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 10 \\ 12 & 7 & 15 & 111 \\ 1 & 0 & 1.2 & 7.4 \end{array}\right]$$

- Row 2 $-$ 12$\times$ row 1: $(0, -5, 3 \mid 111 - 120 = -9)$.
- Row 3 $-$ 1$\times$ row 1: $(0, -1, 0.2 \mid 7.4 - 10 = -2.6)$.
- Row 3 $-$ 0.2$\times$ row 2: $(0, 0, 0.2 - 0.6 = -0.4 \mid -2.6 + 1.8 = -0.8)$.
- Row 3 $\div(-0.4)$: $(0, 0, 1 \mid 2)$.
- Row 2 $-$ 3$\times$ row 3: $(0, -5, 0 \mid -9 - 6 = -15)$; then $\div(-5)$: $(0, 1, 0 \mid 3)$.

- Row 1 − row 2 − row 3: $(1, 0, 0 \mid 10 - 3 - 2 = 5)$.

$$[R \mid d] = \left[\begin{array}{ccc|c} 1 & 0 & 0 & 5 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

3 pivots, 0 free variables, no contradiction. Therefore **exactly one solution**:

$$\boxed{x = \begin{bmatrix} 5 \\ 3 \\ 2 \end{bmatrix}} \qquad \text{₹5L Nifty, ₹3L bonds, ₹2L midcap}$$

Check: $5 + 3 + 2 = 10$ ✓, $60 + 21 + 30 = 111$ ✓, $5 + 0 + 2.4 = 7.4$ ✓.

Example B: infinitely many solutions

Nifty, bonds, balanced, target $(10, 105, 7)$. Concepts 4 and 5 above already reduced it:

$$[R \mid d] = \left[\begin{array}{ccc|c} 1 & 0 & 0.5 & 7 \\ 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

No contradiction ($0 = 0$). But x_3 is free. Therefore **infinitely many solutions**.

Let $x_3 = t$. Then $x_1 + 0.5t = 7$, so $x_1 = 7 - 0.5t$; and $x_2 + 0.5t = 3$, so $x_2 = 3 - 0.5t$:

$$\boxed{x = \begin{bmatrix} 7 - 0.5t \\ 3 - 0.5t \\ t \end{bmatrix}}$$

Check $t = 2$: $(6, 2, 2)$. Budget $6 + 2 + 2 = 10$ ✓, return $72 + 14 + 19 = 105$ ✓, exposure $6 + 0 + 1 = 7$ ✓.

Example C: no solution

The same funds, but target $(10, 105, 5)$. The same row operations act on the new target column: $105 - 120 = -15$; $5 - 10 = -5$; $-5 - 0.2(-15) = -5 + 3 = -2$; $-15/(-5) = 3$; $10 - 3 = 7$.

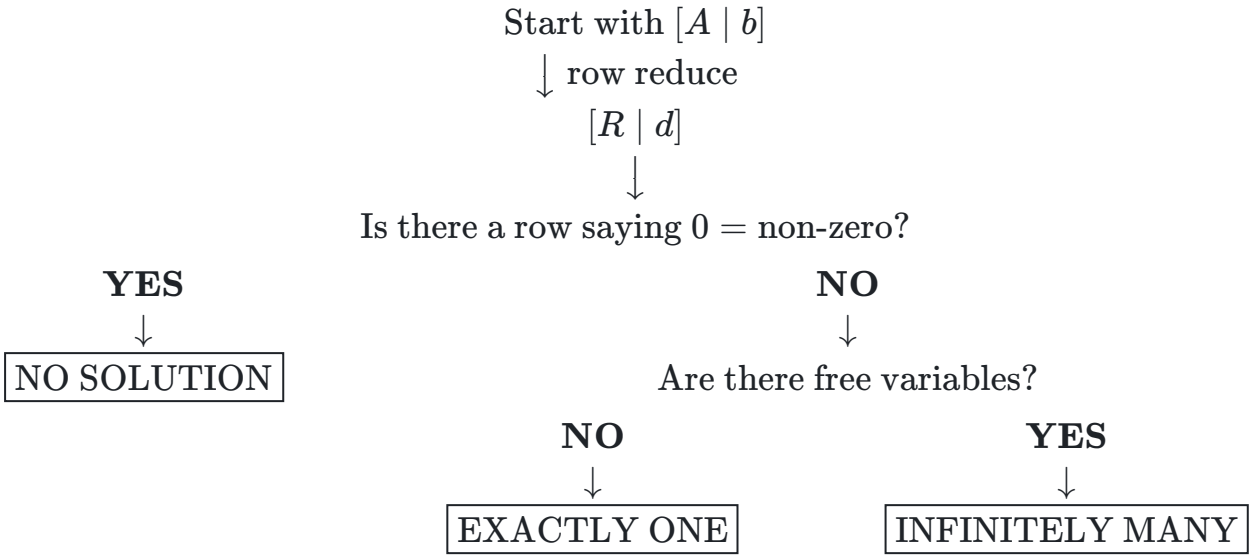
$$[R \mid d] = \left[\begin{array}{ccc|c} 1 & 0 & 0.5 & 7 \\ 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & -2 \end{array}\right]$$

The last row says

$$\boxed{0 = -2}$$

Impossible. Therefore **no solution.** (For these funds, ₹10 lakh earning ₹105 thousand forces exposure 7; asking for 5 contradicts it.)

5. Final decision tree



Contradiction?	Free variables?	Answer	Examples above
yes	(doesn't matter)	no solution	2 × 2 C, 3 × 3 C
no	no	exactly one	2 × 2 A, 3 × 3 A
no	yes	infinitely many	2 × 2 B, 3 × 3 B

Always check for a contradiction **first:** 2 × 2 B and C use the same funds (with a free variable), but in C the contradiction wins.

6. Connection between Ax = 0 and Ax = b

	$Ax = 0$ **(Note 1)**	$Ax = b$ **(Note 2)**
Find	all inputs that produce ZERO	all inputs that produce a SPECIFIC TARGET
Main concepts	null space	particular solution + null space

The complete solution is

$$\boxed{x = x_p + x_n} \qquad \begin{aligned} x_p &= \text{one solution that reaches } b \\ x_n &= \text{any null space solution} \end{aligned}$$

because $Ax_p = b$ and $Ax_n = 0$, so $A(x_p + x_n) = b$.

On the four-fund menu (Note 1's bigger example)

Nifty, ETF, bonds, balanced, target $(10, 105, 7)$. The same five row operations as Note 1, applied to the target column too, give:

$$[\,R \mid d\,] = \left[\begin{array}{cccc|c} 1 & 1 & 0 & 0.5 & 7 \\ 0 & 0 & 1 & 0.5 & 3 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

No contradiction; free variables x_2 (ETF), x_4 (balanced). Free $= 0$ gives $x_p = (7, 0, 3, 0)$. Note 1 gave the null space: $x_n = c\,s_1 + d\,s_2$. So:

$$x = \underbrace{\begin{bmatrix} 7 \\ 0 \\ 3 \\ 0 \end{bmatrix}}_{x_p} + \underbrace{c \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -0.5 \\ 0 \\ -0.5 \\ 1 \end{bmatrix}}_{x_n}$$

Check $c = 3, d = 2$: $(7 - 3 - 1,\ 3,\ 3 - 1,\ 2) = (3, 3, 2, 2)$. Budget $3 + 3 + 2 + 2 = 10$ ✓, return $36 + 36 + 14 + 19 = 105$ ✓, exposure $3 + 3 + 0 + 1 = 7$ ✓.

In money terms: once one portfolio hits the target, swapping Nifty for the ETF or building balanced from Nifty and bonds keeps it hit.

Try it: any target, any free amounts

Budget B 10 Return R 105 Exposure E 7

ETF c 2 Balanced d 2

Zero-row test: $7B + 5E - R = 7(10) + 5(7) - 105 = 0$.

$[\,R \mid d\,]$ has last column $d = (7, 3, 0)$. **One answer** (free amounts = 0): $x_p = (7, 0, 3, 0)$.

Every answer (ETF $= c$, balanced $= d$): $x = x_p + 2\,s_1 + 2\,s_2 = (4, 2, 2, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(4) + 1(2) + 1(2) + 1(2) = 10$ ✓
Return: $12(4) + 12(2) + 7(2) + 9.5(2) = 105$ ✓
Exposure: $1(4) + 1(2) + 0(2) + 0.5(2) = 7$ ✓

Hits the target. Change c and d: it still does.

7. Final memory table

	$Ax = 0$	$Ax = b$
Goal	Find inputs that give ZERO.	Find inputs that give TARGET b.
Row reduce	$A \to R$	$[A \mid b] \to [R \mid d]$
Free variables	Create special solutions.	Decide between one and infinitely many.
Key objects	Special solutions form the null space; nullspace matrix N.	Particular solution x_p (free variables $=$ 0).
Key relationship	$AN = 0$	$x = x_p + x_n$
Contradiction	never ($0 = 0$ always)	$0 =$ non-zero $\to$ no solution
Outcomes	No free variables $\to$ only $x = 0$. Free variables $\to$ infinitely many.	No contradiction + no free variables $\to$ exactly one. No contradiction + free variables $\to$ infinitely many.